How does the tech work (explained by Exponent). Lithium (Li) ions flow from the cathode to the anode when charging a cell and get absorbed. The faster you charge, the more ions flow, crowding around the anode. If unchecked, this crowding leads to lithium plating, resulting in significant cell degradation.

(L to R) Sanjay Byalal, COO, and Arun Vinayak, CEO, Exponent Energy

Exponent solves this by combining their battery management system (10 times more accurate than the industry standard), virtual cell model, and dynamic charging algorithms. This allows them to proactively sense lithium crowding in real time and ensure each cell is rapidly charged without significant degradation.

Further, Exponent has built an advanced HVAC system that is 'off-boarded' from the vehicle to its e^pump. While charging, the e^pump transfers refrigerated water through the e^plug to cool every Li-ion cell in the e^pack. This ensures the e^pack's temperature doesn't exceed 35°C in ambient conditions.

Expansion and raising funds. Focused on commercial electric vehicles for the next 24 months, the startup is sure its franchisee model for setting up these electric charging "pumps" will thrive. There are 30 Exponent pumps set up in Bengaluru (each costing sub-₹500,000), most of which are under the franchise model.

"As a startup, there is no point in putting equity in everything. We are selling the charger via the franchisee model, we are selling the charger with vehicles, we are deploying these for fleet operators, we are working with the finance organisations, so, we have made the entire commercial EV ecosystem work together," Vinayak explains.

This business model around chargers is allowing the startup to save on capital. On the battery side, the company can make 1,000 plus batteries per month and is finalising an electronics contract manufacturer to scale up to three times the current capacity. Its expansion plans include venturing out into five more major Indian cities.

Exponent is backed by Lightspeed India, YourNest VC, 3one4 Capital, AdvantEdge VC, Hero MotoCorp's Chairman and CEO Dr Pawan Munjal's family office, Motherson Group, and has raised $18 million. Reliance-backed electric-three maker Altigreen is one of the prominent customers of the startup.

—Mukul Yudhveer Singh

2 AN AMAZON FOR MANUFACTURING SERVICES

Amazon introduced us to the world of on-demand products. Everything at the click of a button! A startup based in Tamil Nadu has extended this idea to manufacturing services

Founded in 2021 by Dr V. Tamizhinian (Tamizh), Frigate Engineering Services Private Limited is an emerging B2B platform that digitalises manufacturing for its clients worldwide. The startup provides on-demand manufacturing services and prototype development through a supply chain.

With a background in mechatronics, Tamizh started his career as a designer, but yearned to witness the realisation of those designs through manufacturing. So, in 2011, Tamizh started seeking out various small businesses and created prototypes for them, in addition to his regular job at a global original equipment manufacturing (OEM) firm.

Initially, what used to be a side hustle became a full-fledged business after Tamizh recognised the value of creating a marketplace that brought dispersed suppliers together and connected them with potential customers beyond their existing circle. From 2021-22, Tamizh operated the company as a one-person enterprise. "My vision was to address the demand-side challenge by becoming a single point of contact for all outsourced components. I started with machining and expanded into sheet metal fabrication and achieved an asset turnover of 4 to 5 million rupees within the first year," he says.

In 2022, Tamizh onboarded Karthikeyan Prakash, Chandrashekhar C., and Iniyavan Vasanthanas as co-founders to develop a common platform to serve their customers, providing an overview of their ongoing projects with Frigate. Frigate's platform allows the client to track the project's progress at various stages, from initiation to raw material procurement, manufacturing commencement, quality control ini-